

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. (AOC) Examination April 2019

OC 829: Synthetic Dyes and Pigments

Date: 29/04/2019 Day: Monday Time: 01.30 p.m. to 02.00 p.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q-I A Choose the correct answer for the following questions. 10

- 1 The chromophore of indigo dyes is represented as
 - a. X
 - b. H
 - c. Y
 - d. None of these
- 2 Colour of indigo is red in
 - a. vapour phase
 - b. toluene
 - c. water
 - d. polar solvent
- 3 As the number of fused ring increases the colour deepens.
 - a. To yellow
 - b. Less
 - c. More
 - d. Not effected

- 4 The group that enhance the fluorescence.
- Diminoflore
 - Luminophore
 - Auxoflore
 - Fluorophore
- 5 Visible range corresponds to
- below 35.5 Kcal.mole⁻¹
 - 35.5 – 71.0 Kcal.mole⁻¹
 - 71.0 – 150 Kcal.mole⁻¹
 - 150 – 250 Kcal.mole⁻¹
- 6 The λ_{max} of 1,5-diNH₂ AQ dyes is similar to
- 3-NH₂AQ
 - 2-NH₂AQ
 - 1-NH₂AQ
 - 2,2-diNH₂AQ
- 7 The chromophore in methyl red dye:
- azo group
 - azoxy group
 - nitroso group
 - a and b
- 8 Examples for strong chromophoric groups are
- NH₂, NR₂, NHR, OH
 - NO₂, N=O, N=N
 - C=C, C=O, C=S
 - None of the above
- 9 Short chain chromophore have
- e – donating group
 - e – accepting group
 - e – donating and e – accepting group
 - none of these
- 10 Auxochrome
- change the colour of the dye i.e. blue to red
 - increases the intensity of dye
 - decrease the intensity of the dye
 - neither (a) and (c)

Q-I B Draw the structure of the following dyes.

10

1 Naphthol yellow S

2 Methyl red

3 Direct black 38

4 Stilbene azo dyes

5 Rosaniline

6 Crystal violet

7 Rhodamine

8 Acid blue 23

9 Saframine T

10 Meldola's blue

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IV Semester of M. Sc. (AOC) Examination April 2019

OC 829: Synthetic Dyes and Pigments

Date: 29/04/2019 Day: Monday Time: 02.00 p.m. to 04.30 p.m. Maximum Marks: 50

Important Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
 - Make suitable assumptions and draw neat figures wherever required.
 - Use of non-programmable calculator is allowed.
 - Show necessary calculations.
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SECTION-I

Q-II Answer the following. 10

- A** Explain procion dyes and mention the problem with monoreactive dyes. 3
- B** Explain the bathochromacity order of disubstituted AQ dyes. 4
- 1,5 < 1,8 < 1,2
- C** Give the synthesis of Ramazol black B and Procion Brilliant Red M-5B. 3

OR

- C** Mention the characteristics of FBA and give the synthesis of any hair dyes. 3

Q-III Answer the following. 10

- A** (i) What is NIR dyes? Mention its uses. 3
- (ii) Give the general approaches to prepare NIR dyes.
- B** Discuss “Brooker Deviation” giving suitable examples. 3
- C** Write note on short chain chromophore. 4

OR

- C** In dye forming process, explain the formation of quinonediimine and its importance. 4

SECTION-II

Q-IV Answer the following. **10**

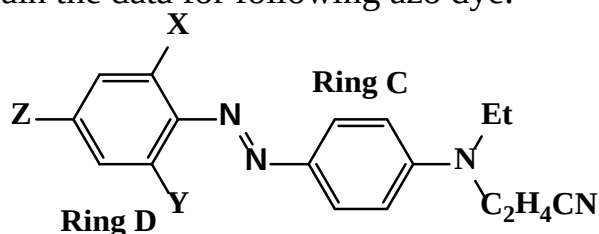
- A** Explain: Chromophore theory giving suitable example. 3
- B** Justify: Heterocyclic diazo compound containing S atom shows more bathochromic effect compared to their benzene analogous. 3
- C** Discuss the tautomerism in case of azo-naphthol and azo phenol dyes. 4

OR

- C** Write note on: Hydroxy azo dyes of benzene series. 4

Q-V Answer the following. **10**

- A** Explain bathochromic effect. 3
- B** Explain the data for following azo dye. 3



Sr. No.	X	Y	Z	$\lambda_{\max}$ (nm)
1)	H	H	NO ₂	453
2)	Cl	H	NO ₂	475
3)	Cl	Cl	NO ₂	417

- C** What is dye? Mention the properties of the dyes. 4

Q-VI Answer the following. **10**

- A** Write note on yellow dyes. 3

OR

- A** Mention the properties of liquid crystalline dyes. 3
- B** Write note on: The image diffusion process. 4
- C** Write note on: Structure and unusual features of Indigo dyes. 3